

Fine-tuning and Deploying a Language Model

The goal of this coding exercise is to build an API endpoint that hosts a fine-tuned Language Model used for *claim detection*, or identifying whether a given sentence contains a factual claim or assertion.¹ The API endpoint should receive a natural language sentence (e.g. “*The Empire State Building is the tallest building in New York City*”) and return a boolean value for whether or not the sentence is a claim, along with a confidence score (e.g. *True, 99%*).

Tasks you could complete:

- Create a high-level diagram of your approach to the exercise, and clearly lay out your “system architecture.” Think through how this endpoint would live in a production environment, and include any details you think may be relevant.
- Fine-tune a Language Model for factual claim detection. [This](#) research paper can be a helpful reference - it also includes information on datasets that you can source for fine-tuning.
- Report and interpret metrics on the performance of your model
- Implement an API endpoint
- Set up robust tests of the API endpoint (e.g. load testing, security features).
- Containerize the deployment
- Anything else you would like to explore related to this problem - tackle this problem in your own way

Important notes:

- **Do not spend an unreasonable amount of time on this exercise—it is perfectly acceptable to only partially complete the exercise.** The scope of this exercise is extensive - set a time limit for yourself (e.g. 10/15/20 hours) and complete as much as you can of the exercise in that time period.
- Completing this project **doesn’t require any hardware beyond what is available on a normal laptop computer** - there is no need for a GPU or external servers for hosting your API endpoint. You should be able to fine-tune the model using only a CPU, and the API only needs to be hosted on localhost. Of course, if you have access to a GPU, you are welcome to use it.
- **Please version control your project using git** (plus host the code on a service like GitHub or GitLab), and make frequent commits as you would in a professional engineering environment - we will review this together afterwards.

¹Note that this task is **not** fact-checking. Claim detection is identifying whether or not a natural language sentence *can be* fact-checked—it is upstream of fact-checking.